Lecture 15: Concentration Inequalities, Weak Law of Large Numbers

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1 Motivation

This week, we'll be investigating probability topics related to the asymptotic behavior of sequences of random variables.

In particular, we're interested in what happens to sums and averages of random variables as the number of random variable summands $n \to \infty$?

1.1 Motivating calculations

Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance σ^2. Define

$$S_n = X_1 + \cdots + X_n, \qquad \bar{X}_n = \frac{S_n}{n}.$$

- $E[S_n] = n\mu$ and $\mathrm{Var}(S_n) = n\sigma^2$: mean and variance gets big as $n \to \infty$, the distribution spreads out!
- $E[\bar{X}_n] = \mu$ and $\mathrm{Var}(\bar{X}_n) = \dfrac{\sigma^2}{n} \to 0$ as $n \to \infty$. The distribution centers around μ tighter and tighter!

This is the mathematical basis for thinking of $E[X]$ as the average of a large number of independent samples.

1.2 "Standardizing" S_n

S_n and $\bar{X}_n$ are at opposite ends of a spectrum (spreading out and shrinking) – what if we consider a different sequence where mean and variance stay constant?

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

- $E[Z_n] = 0$ and $\text{Var}(Z_n) = 1$.
- $S_n - n\mu$ centers the sum around zero; dividing by $\sigma\sqrt{n}$ fixes the variance to 1.

Preview: what does the distribution of Z_n look like? Lots of distributions have mean 0 and variance 1, but they differ in their shape (revealed by PMF/PDF or CDF)! This will be the **central limit theorem (CLT)** which we'll talk about next lecture.

1.2.1 Example: Z_n where $X_i \sim$ Bernoulli(p)

Suppose $X_1, \cdots, X_n$ are i.i.d. Bernoulli(p) and let $S_n = X_1 + \cdots + X_n$.

Let's look at some plots of realizations of samples of $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$ as $n \to \infty$.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

rng = np.random.default_rng(151)

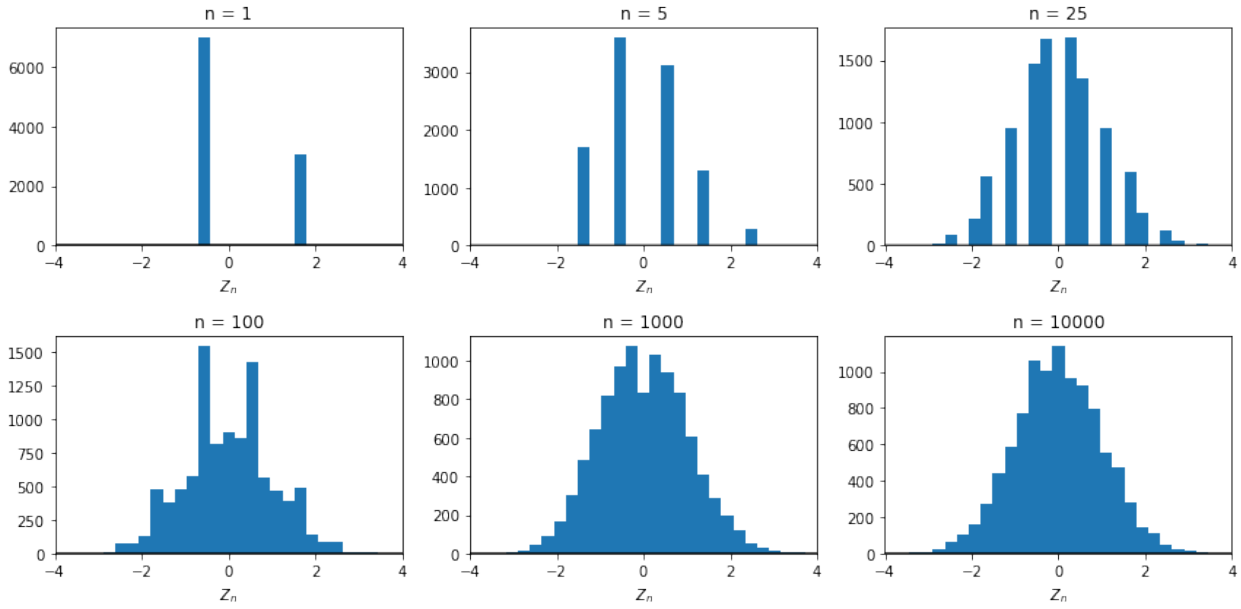
p = 0.3
sigma = np.sqrt(p * (1 - p))
reps = 10_000                          # simulated copies of Z_n for each n
ns = [1, 5, 25, 100, 1000, 10_000]

fig, axes = plt.subplots(2, 3, figsize=(12, 6))
x = np.linspace(-4, 4, 400)

for ax, n in zip(axes.flat, ns):
    # S_n ~ Binomial(n, p)
    S = rng.binomial(n, p, size=reps)
    z = (S - n * p) / (sigma * np.sqrt(n))       # Z_n = (S_n - np) / (sigma sqrt(n))

    ax.hist(z, bins=np.linspace(-4, 4, 30), density=False)
    ax.plot(x, norm.pdf(x), "k")
    ax.set_xlim(-4, 4)
    ax.set_title(f"n = {n}")
    ax.set_xlabel(r"$Z_n$")

plt.tight_layout()
plt.show()
```



2 Concentration Inequalities or Tail Bounds

> **Definition**
>
> A **concentration inequality** (or **tail bound**) is an upper bound on the probability that a random variable occurs far from its mean.

2.1 Markov's Inequality

The most fundamental tail bound is Markov's inequality.

> **Theorem** (Markov)
>
> If X can take only nonnegative values and $a > 0$, then
>
> $$P(X \geq a) \leq \frac{E[X]}{a}.$$

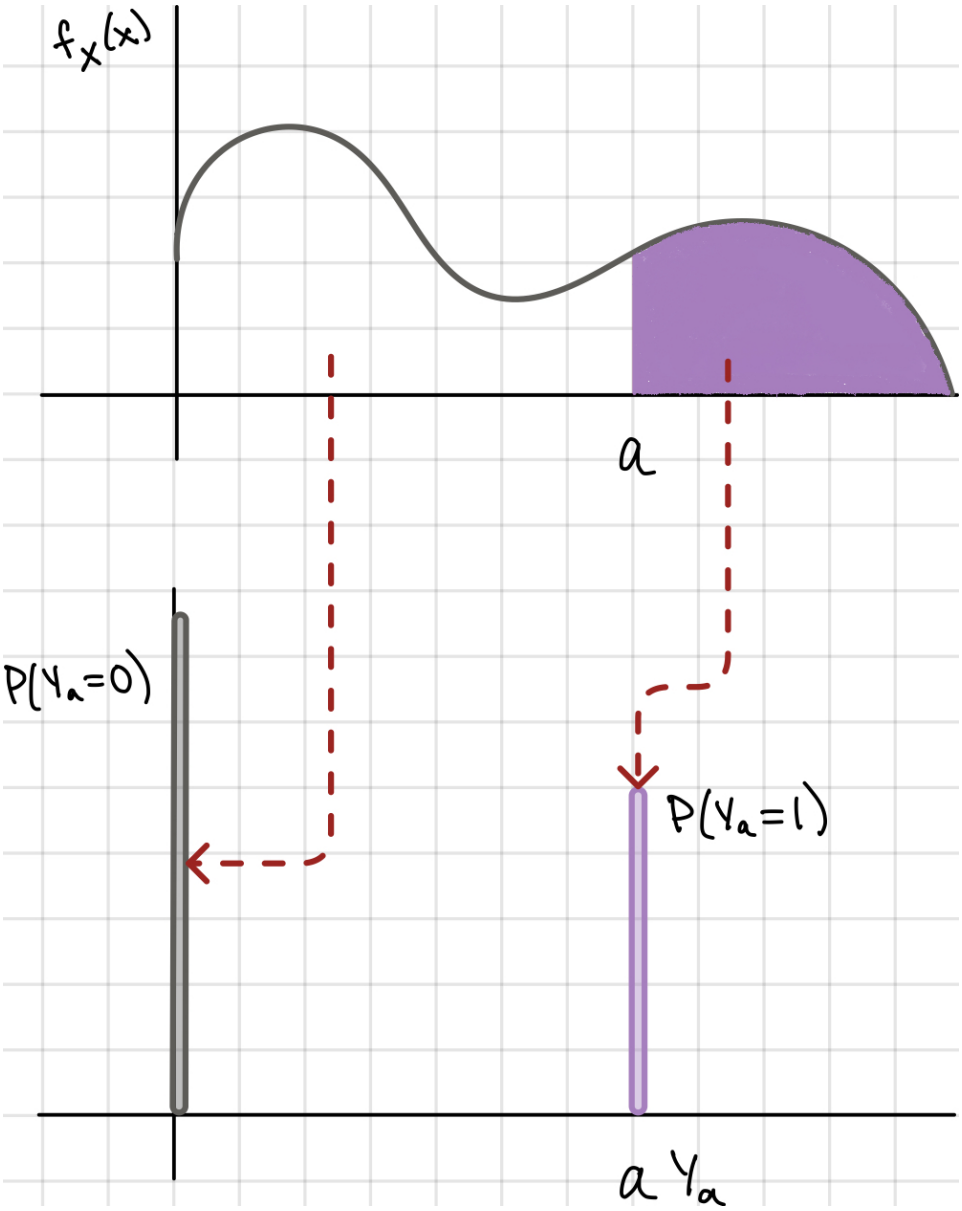
Proof. Define Y_a to be an indicator random variable,

$$Y_a = \begin{cases} 0 & \text{if } X < a \\ 1 & \text{if } X \geq a \end{cases}.$$

Since $X \geq 0$, we have $a \cdot Y_a \leq X$ pointwise. (Check the two cases!)

Taking expectations gives $a\,P(X \geq a) = a\,E[Y_a] \leq E[X]$. $\square$

2.1.1 Visualization: Markov's Inequality



Since probability mass is shifted to the left, the expectation can only decrease!

2.1.2 Exercise

> **Exercise**
>
> Let $X \sim \text{Exp}(1)$ be an exponential random variable. Use Markov's inequality to bound $P(X \geq a)$ and compare with the exact value.

Solution

Since $X \sim \text{Exp}(1)$, $X \geq 0$ and $E[X] = 1$. Markov's inequality gives, for $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a} = \frac{1}{a}.$$

The exact value is

$$P(X \geq a) = \int_a^\infty e^{-x}\, dx = e^{-a}.$$

---

a	Markov bound $1/a$	Exact e^{-a}
1	1	0.368
2	0.5	0.135
5	0.2	0.0067
10	0.1	0.0000454

Markov is valid but loose: it decays like $1/a$, while the true tail decays exponentially. (For $a \leq 1$ the bound is trivial.)

2.2 Chebyshev's Inequality

Now, we turn to a slightly stronger concentration inequality. Note that this applies to random variables with finite mean and variance!

Theorem (Chebyshev)

If X is a random variable with mean μ and variance σ^2, then for $c > 0$

$$P(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}.$$

Proof. Let $Y = (X - \mu)^2$. Then $Y \geq 0$, so Markov's inequality applies to Y.

For $c > 0$, these events are the same:

$$|X - \mu| \geq c \iff (X - \mu)^2 \geq c^2.$$

Apply Markov to Y with $a = c^2$:

$$P(|X - \mu| \geq c) = P(Y \geq c^2) \leq \frac{E[Y]}{c^2} = \frac{E[(X - \mu)^2]}{c^2} = \frac{\sigma^2}{c^2}. \qquad \square$$

Alternative form

Take $c = k\sigma$:

$$P(|X - \mu| \geq k\sigma) \leq \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}.$$

2.2.1 Exercise

Exercise

Let $X \sim \text{Exp}(1)$. Use Chebyshev's inequality to bound $P(|X - 1| \geq c)$ and compare with the exact value.

Solution

For $X \sim \text{Exp}(1)$, $\mu = E[X] = 1$ and $\sigma^2 = \text{Var}(X) = 1$. Chebyshev's inequality gives, for $c > 0$,

$$P(|X - 1| \geq c) \leq \frac{\sigma^2}{c^2} = \frac{1}{c^2}.$$

Exact value. The event $|X - 1| \geq c$ means $X \geq 1 + c$ or $X \leq 1 - c$. For $c > 1$, the second is impossible since $X \geq 0$, so

$$P(|X - 1| \geq c) = P(X \geq 1 + c) = e^{-(1+c)}.$$

(For $0 < c \leq 1$ there is an extra term: $e^{-(1+c)} + 1 - e^{-(1-c)}$.)

---

c	Chebyshev bound $1/c^2$	Exact
1	1	0.135
2	0.25	0.0498
3	0.111	0.0183
5	0.04	0.00248

Chebyshev is valid but loose: it decays like $1/c^2$, while the true tail decays exponentially. (For $c \leq 1$ the bound is trivial.)

3 Weak Law of Large Numbers (WLLN)

Now we turn to the Weak Law of Large Numbers, which answers the question: How often does the sample mean $\bar{X}_n$ stay near μ?

3.1 Application to $\bar{X}_n$ where $X_i \sim$ Bernoulli(p)

Markov: $P(\bar{X}_n \geq a) \leq p/a$, which does not shrink with n.

The variance of $\bar{X}_n$ is going to 0 as $n \to \infty$, so we should be able to do better than a constant tail bound probability! The answer is to apply Chebyshev's inequality (if the random variables have finite variance).

3.2 Weak Law of Large Numbers

Theorem (WLLN)

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ. For every $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \to 0 \quad \text{as } n \to \infty.$$

Proof (for case where Var$(X_i) = \sigma^2$). Assume $\sigma^2 < \infty$. Fix $\varepsilon > 0$.

By linearity of expectation and independence,

$$E[\bar{X}_n] = \mu, \qquad \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^{n} \text{Var}(X_i) = \frac{\sigma^2}{n}.$$

Apply Chebyshev to $\bar{X}_n$ with $c = \varepsilon$:

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

The right side goes to 0 as $n \to \infty$ for fixed $\varepsilon > 0$, so $P(|\bar{X}_n - \mu| \geq \varepsilon) \to 0$.

Intuition. For large n, $\bar{X}_n$ lies in the interval $(\mu - \varepsilon, \mu + \varepsilon)$ with probability close to 1, however small ε is. Smaller ε just requires larger values of n!

3.2.1 Example: polling

Suppose we want to estimate the fraction p of voters supporting a candidate. How many people must we poll so the estimate is within 0.01 of p with probability at least 0.95?

Interview n "randomly selected" voters and let X_i be an indicator random variable that indicates if voter i supports the candidate. Then $\bar{X}_n$ is the fraction of the n voters supporting the candidate.

Note that the $X_i \sim \text{Bernoulli}(p)$ are independent, so Chebyshev yields

$$P(|\bar{X}_n - p| \geq \epsilon) \leq \frac{p(1-p)}{n\epsilon^2} \leq \frac{1}{4n\epsilon^2}$$

.

Now, if we want the gap $\epsilon = 0.01$ and the probability

$$P(|\bar{X}_n - p| \geq \epsilon) \leq 1 - 0.95 = 0.05,$$

then we can achieve this by choosing n such that the upper bound is less than 0.05,

$$\frac{1}{4n(0.01)^2} \leq 0.05$$

or $n \geq 50000$.

3.2.2 Example: Cauchy RV sample mean

A **standard Cauchy** random variable has PDF

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \qquad x \in \mathbb{R}.$$

It is symmetric and bell-shaped like the normal, but its tails are much heavier:

$$P(|X| \geq a) = \frac{2}{\pi} \arctan\left(\frac{1}{a}\right) \approx \frac{2}{\pi a} \quad \text{for large } a.$$

(It is also the ratio of two independent $N(0,1)$ random variables.)

If the X_i are standard Cauchy, $\bar{X}_n$ is again standard Cauchy for every n: it never concentrates.

Why doesn't the WLLN apply? The expectation does not exist:

$$E|X| = 2\int_0^\infty \frac{x}{\pi(1+x^2)}\,dx = \frac{1}{\pi}\Big[\ln(1+x^2)\Big]_0^\infty = \infty.$$

The positive part of X contributes $+\infty$ and the negative part $-\infty$, so $E[X]$ is undefined.

Markov's inequality applied to $|X|$ gives

$$P(|X| \geq a) \leq \frac{E|X|}{a} = \infty,$$

which says nothing. Chebyshev needs a finite variance, so it fails too.



Warning

The WLLN fails here because $E[X]$ does not exist: Markov (and hence Chebyshev) gives no information.